

MAJOR VESSELS

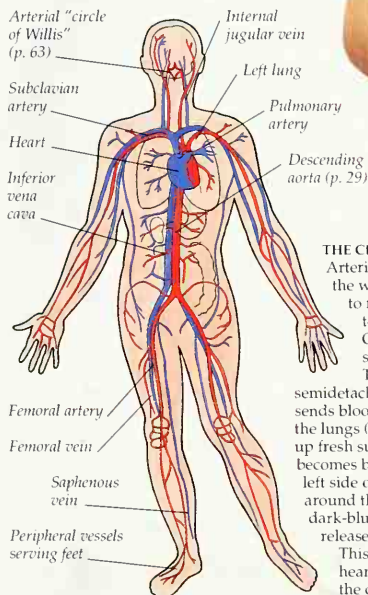
Arteries and veins have similar layers in their walls, but in different proportions. The arterial wall is thick, muscular, and elastic to withstand the bulging surges of high-pressure blood squirting from the heart. The vein wall is loose and stretchy, as its blood is under very low pressure, oozing back to the heart. Valves prevent blood flowing in the wrong direction.

Zygomatic muscle tendon pulls on lip to sneer

Branches of facial vein

VESSELS IN THE FACE AND NECK

The neck is a major thoroughway for tubes connecting the head and torso. In addition to the windpipe and gullet, several main blood vessels service the brain, face, and sense organs. This model chiefly displays the veins, in particular the large internal jugular vein that brings dark, low-oxygen blood from the brain, skull, and much of the face.



THE CIRCULATORY SYSTEM

Arteries snake out through the whole body, dividing to reach every part, down to the fingertips and toes. Only arteries from the left side of the heart do this. The heart is really two semidetached pumps. The right one sends blood on a shorter journey to the lungs (pp. 26-27). Here it picks up fresh supplies of oxygen and becomes bright red. It returns to the left side of the heart, gets pumped around the body, and becomes dark-blush red in color as it releases its oxygen to the tissues. This blood returns to the heart's right side to complete the double circulation.

Frontal branches of superficial temporal artery

Top of skull removed to show brain

Cerebral arteries and veins to brain (pp. 60-61)

Parotid (salivary) gland (p. 34)

Lingual vein

Windpipe (pp. 24-25)

Internal jugular vein

First rib (pp. 14-15)

Subclavian vein

External jugular vein

Subclavian artery